Arithmetic formulas for Gaussian-controlled tanh feature paths

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Abstract

We study globally margined Boolean classifiers represented by Gaussian variational feature paths with tanh activation. The Gaussian controls may have nonzero means, arbitrary covariances, varying query diagonals, and unrestricted real precision. If their relative entropies are $D_1, \dots, D_H$, the classifier has rational arithmetic sign formulas, followed by a majority, with resources polynomial in

$$(d + H + \|v\| + 2) \exp\left[C \sum_{j=1}^{H} (\sqrt{D_j} + D_j) \right].$$

Consequently logarithmic entropy occupation gives nonuniform PNC^1 circuits, hence logarithmic space with polynomial advice. The construction uses normalized Gaussian Hermite features, degree-dependent averaging, rare nonlinear branches, and an artificial variance floor that removes hidden assumptions about the precision of importance coefficients. The Gaussian scalar baseline is strictly larger than the unrestricted one. Near-baseline consequences are therefore stated relative to the Gaussian class's own baseline. This is a theorem about Gaussian variational controls. The unrestricted critical Bayesian posterior permits non-Gaussian row tilts; no replacement of those tilts by Gaussian controls is proved here.

1 Model and statement

Let $d \geq 1$, $H \geq 1$, $X_d = \{-1, 1\}^d$, and $h_0(x) = (1, x)/\sqrt{d}$. A feature state is a map from X_d into a real separable Hilbert space. At stage j, choose a Gaussian row law with mean m_j and covariance $\Sigma_j > 0$, relative to the standard isonormal Gaussian on the preceding feature span. The next kernel is

$$Q_j(x, y) = \mathbb{E}\big[\tanh(\langle G_j, h_{j-1}(x)\rangle) \tanh(\langle G_j, h_{j-1}(y)\rangle)\big], \qquad G_j \sim N(m_j, \Sigma_j). \tag{1}$$

Any Hilbert feature realization of this kernel may be used in the next stage. The control cost is

$$D_j = \frac{1}{2}\left[\|m_j\|^2 + \operatorname{tr}(\Sigma_j - I - \log \Sigma_j) \right]. \tag{2}$$

Only the finite span of the 2^d query states matters. We extend a control by mean zero and covariance identity on its orthogonal complement. This preserves both the query law and its relative entropy. Coordinates in the resulting fixed countable feature bases may be arbitrary real numbers; they need not be sparsely supported or bounded away from zero.

The final output is $z(x) = \langle v, h_H(x)\rangle$. Throughout the main statement, $f(x)z(x) \geq 1$ on the cube. A different positive margin is handled by scaling v.

Theorem 1.1. *There are universal constants C, c with the following property. Every Gaussian-controlled path above has a classifier that is a majority of signs of rational arithmetic formulas*

whose total size and coefficient bit lengths are at most R^c, where

$$R = (d + H + \|v\| + 2) \exp\left[C \sum_j (\sqrt{D_j} + D_j)\right].$$

Their majority agrees with f on every Boolean input. In particular, whenever R is polynomial in an ambient input resource, f belongs to nonuniform PNC^1*, and therefore to* L/poly.

The arithmetic-formula conclusion is stronger and more precise than a claim about a real arithmetic computation with shared intermediate states. All expressions below are scalar trees. A repeated occurrence of an already sampled child is charged its full size; distinct child calls are independent. Random choices select an expression independently of the queried input. No algorithm for finding the nonuniform constants or for sampling them efficiently is asserted. Identically zero controls or corrections are omitted. In particular, a zero branch probability means that no correction is constructed and no division by that probability is performed.

2 Scalar-oracle calculus

An oracle family $X_i(x)$ has mean vector $u(x)$, pointwise trace moment $\sum_i \mathbb{E}X_i(x)^2 \le V$, and common expected scalar-tree size at most T. We charge $T \ge 1$. The coordinate index set may be countable. Trace bounds are pointwise in x; we do not interchange a sum with a supremum over x.

Lemma 2.1 (Linear and quadratic sampling)**.** *For a Hilbert–Schmidt matrix K, each coordinate of Ku has an unbiased scalar oracle of expected size $O(T)$, with the sum of its second moments at most $\|K\|_{\mathrm{HS}}^2 V$. For a Hilbert–Schmidt matrix B, the scalar $u^T B u$ has an unbiased oracle of expected size $O(T)$ and second moment at most $\|B\|_{\mathrm{HS}}^2 V^2$.*

Proof. In row i, put $b_i = \sum_j K_{ij}^2$, sample j with probability K_{ij}^2/b_i, and return $K_{ij}X_j$ divided by that probability. Its second moment is $b_i \sum_{j:K_{ij}\neq 0} \mathbb{E}X_j^2 \le b_i V$. For the quadratic form, sample (i,j) with probability $B_{ij}^2/\|B\|_{\mathrm{HS}}^2$ and use independent children $X_i X_j'$. Tonelli's theorem gives the claimed bound, including countable bases. □

For $A = J + K$, where J is a signed coordinate injection, retain the copied child and add the sampled K child only on a Bernoulli branch of probability $p = \|K\|_{\mathrm{HS}}$ when this norm is small. Divide the correction by p. Independence gives

$$V_{\mathrm{new}} \le V + 3\|K\|_{\mathrm{HS}} V, \qquad T_{\mathrm{new}} \le T(1 + C\|K\|_{\mathrm{HS}}). \tag{3}$$

Indeed the cross term is at most $2\|K\|_{\mathrm{HS}}\|u\|^2$ and the new second-moment term is $\|K\|_{\mathrm{HS}}^2 V/p$. On the absent branch we return the copied formula. Fixed scalar weights are merged along copy chains. For large $\|K\|_{\mathrm{HS}}$, copy J and sample only K, then average polynomially many copies when the true output mean norm is bounded. Sampling the whole matrix $J + K$ would not be justified in a countable basis.

An average of two independent oracle families has trace moment

$$\frac{1}{2}\left(V + \|u(x)\|^2\right). \tag{4}$$

Thus a bounded-mean family whose moment bound increases by $O(\eta_j)$ per small step can be kept at a fixed bound by $O(1 + \sum_j \eta_j)$ two-copy resets. Their cost is $\exp[O(\sum_j \eta_j)]$.

Lemma 2.2 (Fixed-dimensional smooth scalar factory)**.** *Let $r \in \mathbb{R}^p$, $p \le 2$, have scalar coordinate oracles of uniformly bounded second moment and expected size T. For any compactly supported $F \in H^4(\mathbb{R}^p)$, there is an unbiased real scalar oracle for $F(r)$ with second moment at most $C\,\|F\|_{H^4}^2$ and expected size at most CT. The same statement holds for a countable family, with weighted sums of the second moments bounded by the corresponding weighted sums of H^4 norms.*

Proof. Use a fixed frequency density proportional to $(1+\|\omega\|^2)^{-4}$. Construct $W = \sum_{l=1}^{p} \omega_l X_l$ from independent coordinate children; its second moment is at most $C\,\|\omega\|^2$. Choose $K \sim \mathrm{Pois}(\lambda)$ with $\lambda = 1 + C\,\|\omega\|^2$, and form

$$Z = \prod_{a=1}^{K} (1 + iW_a/\lambda).$$

The Poisson product identities give $\mathbb{E}Z = e^{i\omega\cdot r}$ and $\mathbb{E}|Z|^2 \le e$. Multiply by the Fourier coefficient divided by the frequency density and take its real part. Plancherel gives the moment bound. A balanced real/imaginary expansion uses $O((K+1)^2)$ child occurrences. The fourth frequency moment is finite in dimensions at most two for the displayed density, so expected real-tree cost is $O(T)$. Copies repeated in that expansion use the same already sampled child; they are not independent redraws. Fixed compact-interval extensions are used whenever F is specified only on a bounded rectangle. □

3 Normalized noncentral Gaussian features

For $b \in \mathbb{R}$, put

$$Y_b = \frac{Z+b}{\sqrt{1+b^2}}, \quad Z \sim N(0,1), \qquad F(t,b) = \mathbb{E}\tanh(\sqrt{t}\,Y_b)^2.$$

Then $\mathbb{E}Y_b^2 = 1$. Expand the normalized scalar feature in orthonormal probabilists' Hermite polynomials:

$$\frac{\tanh(\sqrt{t}\,Y_b)}{\sqrt{F(t,b)}} = \sum_{k\ge 0} c_k(t,b) H_k(Z). \tag{5}$$

At $t = 0$ the continuous extension has

$$c_0(0,b) = \frac{b}{\sqrt{1+b^2}}, \qquad c_1(0,b) = \frac{1}{\sqrt{1+b^2}}, \qquad c_k(0,b) = 0 \ (k \ge 2).$$

For a unit vector u, the feature map

$$\mathcal{F}_{t,b}(u) = \bigoplus_{k\ge 0} c_k(t,b) u^{\otimes k} \tag{6}$$

has norm one. Using full ordered tensor powers makes its coordinate description explicit. Its inner products give exactly the normalized Gaussian tanh kernel, including different (t,b) at different queries.

Lemma 3.1 (Coefficient and variance bounds)**.** *For all $t > 0$ and $b \in \mathbb{R}$,*

$$\sum_{k\ge 2} k^2 c_k(t,b)^2 \le 8t^2. \tag{7}$$

Mixed derivatives in t, b of every fixed order (in particular through order five), with two further derivatives in Z, of the normalized scalar feature in (5) are bounded in Gaussian L^2 for $0 \le t \le 1$, uniformly in b. On larger bounded parameter rectangles the bounds are polynomial in their ranges. Moreover

$$F(t,b) \le g(t) := \min\{1, t/(1+t/2)\}. \tag{8}$$

Proof. The Gaussian second-derivative identity and $k^2 \le 2k(k-1)$ for $k \ge 2$ give

$$\sum_{k\ge 2} k^2 c_k^2 \le \frac{2t^2}{(1+b^2)^2 F(t,b)} \mathbb{E}\tanh''(\sqrt{t}Y_b)^2 \le 8t^2,$$

since $\tanh''(z)^2 \le 4\tanh(z)^2$. For the derivative assertion, write

$$\tanh(\sqrt{t}Y_b)/\sqrt{t} = Y_b \psi(tY_b^2), \qquad \psi(s) = \tanh(\sqrt{s})/\sqrt{s}.$$

Every fixed derivative of ψ is bounded on $[0,\infty)$; the Taylor expansion handles zero and direct differentiation handles infinity. Every fixed b derivative of Y_b is polynomially bounded in $|Z|$, uniformly in b. Also $F(t,b)/t$ is smooth and bounded away from zero for $0 \le t \le 1$, uniformly in b: $\mathbb{E}Y_b^2 = 1$ and uniformly bounded fourth moments give a uniform small-ball lower bound. Differentiation under the Gaussian integral now proves the stated bounds. For $t \ge 1$, the same small-ball argument bounds $F(t,b)$ below by a positive constant and direct differentiation gives polynomial bounds. Finally, $\tanh^2 z \le z^2/(1+z^2/2)$. For example, $w(z) = z/\sqrt{1+z^2/2}$ satisfies $w(0) = 0$ and $w'(z) \ge 1 - w(z)^2$ for $z \ge 0$, since $(1+s)^{-1/2} \ge 1 - s$ for $s \ge 0$; scalar comparison with the tanh equation proves the inequality. Jensen's inequality for the concave function $s/(1+s/2)$ and $\mathbb{E}Y_b^2 = 1$ prove (8). □

In particular, on a fixed bounded rectangle in (r,ζ), for small a and bounded M, the functions $c_k(ar, M\zeta)$ have weighted H^4 tail norm

$$\sum_{k\ge 2} k^2 \left\| c_k(a\,\cdot, M\,\cdot)\right\|_{H^4}^2 \le Ca^2. \tag{9}$$

The two low-degree differences from $c_k(0, M\zeta)$ have H^4 norm $O(a)$, and

$$\left\| F(ar, M\zeta)/g(a) - r \right\|_{H^4} \le Ca. \tag{10}$$

These follow from Lemma 3.1, scaling each t derivative by a, and the fact that the differences vanish at $a = 0$. For large parameter ranges, replace the right sides by fixed polynomials in those ranges. A bounded extension operator into compactly supported H^4 functions preserves these weighted estimates.

4 The path compiler

We first describe a deterministic-operator version. Let each linear covariance map be $A_j = J_j + K_j$, with

$$\kappa_j = \|K_j\|_{\mathrm{HS}}, \quad M_j = \|m_j\|, \quad \rho_j = \min\{1, \sigma_{\min}(A_j)\}, \quad S = \sum_j (\kappa_j + M_j), \quad L = \sum_j \log(1/\rho_j).$$

The compiler has expected scalar-tree cost

$$\mathrm{poly}(H + T_0)\exp[C(S+L)] \tag{11}$$

for normalized feature directions. We maintain a unit direction $u(x)$ and a scalar $r(x) \in (0,1]$ such that

$$h(x) = \sqrt{\bar{q}\, r(x)}\, u(x),$$

where $\bar{q}$ is deterministic. The combined mean squared norm is at most two. Common trace moments are kept at a fixed bound using (4). Initially $r = 1$ and the normalized cube coordinates have bounded deterministic formulas.

4.1 Control and mean coordinates

At one stage put

$$n = \|Au\|, \qquad \beta = \langle m, u\rangle, \qquad u_G = Au/n, \qquad b = \beta/n, \qquad c = (1+\kappa)^2 + M^2.$$

The total preactivation second moment is $\bar{q}r(n^2+\beta^2)$. Set

$$a = c\bar{q}, \qquad r' = r(n^2+\beta^2)/c. \tag{12}$$

The row at query x has the distribution $\sqrt{ar'(x)}Y_{b(x)}$, with Gaussian direction $u_G(x)$. Its leading unit feature at variance zero is

$$\ell(x) = \frac{(\beta, Au)}{\sqrt{n^2+\beta^2}} = \frac{(b, u_G)}{\sqrt{1+b^2}}. \tag{13}$$

For a small joint control $\eta = \kappa + M$, the squared-norm corrections are quadratic forms with Hilbert–Schmidt norm $O(\eta)$:

$$A^*A - I = J^*K + K^*J + K^*K, \qquad A^*A + mm^* - I = (A^*A - I) + mm^*.$$

Lemma 2.1 supplies $O(\eta^2)$-second-moment oracles for these corrections. Apply Lemma 2.2 after division by η to the smooth function $(1+\eta s)^{-1/2} - 1$, on a fixed interval where its denominator is bounded away from zero. Its H^4 norm is $O(\eta)$. Adding the resulting normalization products on probability-η branches constructs u_G and ℓ with moment increase and multiplicative common cost $O(\eta)$. For the new degree-zero component of ℓ, create the β oracle only on a probability-η branch and divide it by η there. Its trace contribution is $O(M^2/\eta) = O(\eta)$ and its expected coordinate cost is $1 + O(\eta T)$. This avoids a fixed per-layer cost factor for a newly appended scalar coordinate. The relative scalar in (12) is updated in the same way, using the quadratic form for $n^2 + \beta^2 - 1$.

When $M > 0$, a transient scalar helper $\zeta = b/M$ has bounded second moment and cost CT: sample $\langle m, u\rangle/M$ and multiply by the normalization oracle for $1/n$. It is used only on subsequent rare nonlinear branches; it is not stored as a new common-cost state that would introduce a fixed multiplicative factor at each layer. For $M = 0$, set $b = 0$ without a helper. The direction u_G and leading feature ℓ are alternative local oracle families. Their trace budgets are not added as if both were one persistent state; only ℓ is retained as the low-degree output base, while u_G is called on the charged correction and tensor branches.

For a joint control bounded away from zero, copy J and sample K, then use the lower bound $n \geq \rho$. Smooth compact extensions of the inverse-square-root functions have Sobolev norms polynomial in $1+\kappa$, $1+M$, and $1/\rho$. Scalar factories, products, and polynomial averaging produce the same exact means at a cost polynomial in these quantities. Its logarithm is $O(\kappa + M + \log(1/\rho))$. The auxiliary b range is bounded by M/ρ and its rescaling is charged once to this joint stage.

4.2 Gaussian features without a tensor variance explosion

The exact new normalized direction has coordinates $c_k(ar', b)u_G^{\otimes k}$. For $k \geq 2$, average $m_k = \lceil kV \rceil$ independent copies of each direction-coordinate oracle before multiplying. Since $\|u_G\| = 1$, their second moments $W_i(x)$ satisfy

$$\left(\sum_i W_i(x)\right)^k \leq \left(1 + \frac{V-1}{m_k}\right)^k \leq e. \tag{14}$$

Estimate the scalar coefficient independently by Lemma 2.2. For small a, create this coordinate only with probability $p_k = a/k^2$, dividing the product by p_k on that branch. Equation (9) and (14) give total tail moment $O(a)$; the common expected coordinate cost is

$$1 + Cp_k(km_kT + T) \le T(1 + Ca).$$

At degrees zero and one, retain the leading feature (13) and add independent probability-a product corrections for $c_k(ar', b) - c_k(0, b)$. Their second moments are $O(a^2)$ and their moment and cost increases are $O(a)$. The summed leading cross term is nonpositive, since $c_0(0, b)[c_0(t, b) - c_0(0, b)] + c_1(0, b)[c_1(t, b) - c_1(0, b)] = \langle Y_b, \phi_{t,b}\rangle - 1 \le 0$ for the two unit Gaussian functions in (5). The scalar r is updated using (10) on another probability-a branch. Its new mean is

$$\bar{q}_{\text{new}} = g(a), \qquad r_{\text{new}} = F(ar', b)/g(a) \in (0, 1]. \tag{15}$$

Thus every small free step costs $1 + O(a)$ and has an $O(a)$ reset charge.

For large a, use $p_k = \min(1, a/k^2)$, the polynomial Sobolev bounds, and polynomially many averages. Include the two low degrees and the scalar variance map directly. This costs polynomially in $1 + a$ and in the parameters of a jointly large control, and restores the combined trace bound. The logarithmic free-step charge is therefore $O(a)$ near zero and $O(\log(1 + a))$ away from zero.

4.3 Deterministic variance telescoping

Put $\ell(a) = \log(a/g(a))$. This is comparable to a near zero and to $\log(1 + a)$ away from zero. Equations (12) and (15) give

$$\sum_{j=1}^{H} \ell(a_j) = \log(\bar{q}_0/\bar{q}_H) + \sum_j \log c_j.$$

Since $c_j \ge 1$ and g is increasing, the deterministic upper variances cool no faster than iteration of g. After the first step they are at most one, and $g(q) = q/(1 + q/2)$ on $(0, 1]$. Hence

$$\bar{q}_H \ge c/(H + 1), \qquad \sum_j \ell(a_j) \le C \log(H + 1) + 2\sum_j \log(1 + \kappa_j + M_j).$$

The small-step moment resets and large-step averaging now prove (11). All transient helpers are called only within the explicitly charged branches. No vector subcomputation is shared.

5 Finite precision without assuming control precision

We give the precision argument because counting real arithmetic operations alone would not prove Theorem 1.1. Let $\overline{V}_i$ denote the cube average of an oracle's second moment. Remove a zero-$\overline{V}_i$ oracle; it is zero almost surely at every cube input. Otherwise inject an independent Rademacher sign:

$$X_i^+ = X_i + \sqrt{\delta \overline{V}_i}\, \varepsilon_i, \qquad \operatorname{Var} X_i^+(x) \ge \gamma \overline{V}_i^+, \quad \gamma = \delta/(1 + \delta). \tag{16}$$

Choose inverse-polynomial δ, inject initially and at stored intermediates, and restore the floor after averaging resets. Means are unchanged, trace moments increase by only $O(H\delta)$, and scalar cost has only an additive polynomial overhead before the previous exponential budget. Conditional versions of the same floor are used inside scalar factories below.

Normalize each oracle subtree by its own cube RMS $a_i = \sqrt{\overline{V}_i}$. These individual scales can be arbitrarily small. We place only the resulting coefficient ratios into the fixed arithmetic tree. The injected-noise node has normalized coefficients $1/\sqrt{1 + \delta}$ and $\sqrt{\delta/(1 + \delta)}$.

Linear and sampled product nodes. For a near-copy node $Y = aX_0 + (B/p)C_J X_J$ with independent branches, the noise floors imply

$$\overline{V}_Y \geq \gamma \left(a^2 \overline{V}_0 + p^{-1} \sum_j p_j C_j^2 \overline{V}_j \right).$$

Its copied normalized coefficient is at most $\gamma^{-1/2}$, and its selected normalized coefficient has squared expectation at most γ^{-1}. For an independent product, the pointwise second-moment floors multiply. An m-term averaged child has a floor at least γ/m relative to its own cube RMS. Hence a sampled degree-k product has normalized coefficient bounded, on an active branch, by $p^{-1/2}(m/\gamma)^{k/2}$. Equivalently, its selected squared coefficient multiplied by the product of the child floors has expectation at most one. This handles quadratic entry sampling and the independently estimated coefficient times a tensor product as well.

Fourier factory nodes. Condition on frequency and Poisson count $\theta = (\omega, K)$ in Lemma 2.2. Let Z be its complex product and put $s_\theta^2 = \mathbb{E}_x \mathbb{E}(|Z|^2 \mid \theta)$. If c denotes its Fourier importance coefficient, replace its conditional real output by

$$Y_\theta = \Re(cZ) + \sqrt{\delta}\,|c| s_\theta \varepsilon.$$

The added noise charges at most δ times the same Fourier moment integral, and the mixed output has a pointwise variance floor. Its conditional RMS b_θ satisfies $b_\theta^2 \geq \delta |c|^2 s_\theta^2$. Normalize each complex factor by $t_j = \sqrt{1 + r_j^2 a_j^2}$, where r_j is its fixed multiplier of the raw child X_j. The normalized factor coefficients are at most one in magnitude, and the child floors give

$$s_\theta^2 \geq \gamma^K \prod_j t_j^2.$$

Thus the conditional output divided by b_θ has root coefficient at most $\delta^{-1/2}\gamma^{-K/2}$, plus a noise coefficient at most one. The mixture normalization contributes b_θ/b, whose squared expectation is exactly one. This removes possible phase cancellation without a lower bound on a Fourier coefficient or a frequency cutoff. For vector frequencies, use the factor $1 + i\sum_l \omega_l X_l/\lambda$ and normalize it by $t^2 = 1 + \sum_l \omega_l^2 a_l^2/\lambda^2$. Its at most three coefficients are bounded by one. Independent coordinate children give a noise variance at least $\gamma \sum_l \omega_l^2 a_l^2/\lambda^2$, so the same factorwise lower bound and product proof apply even when their means cancel.

Fixing and rounding a good seed. Sample the final linear readout with probabilities proportional to its squared coordinates. It has second moment at most a constant times $\|v\|^2$. Average polynomially many copies per group, and use $O(d+1)$ independent groups. Chebyshev, a Chernoff bound, and a union bound over the cube give a realization in which, at every input, at least three quarters of the group averages are within $1/2$ of the target output. Those groups have the correct signs and absolute margin at least $1/2$; the other groups need not have any margin. The expected total formula size and the expected number of primitive calls are polynomial in R.

A size Markov bound and the normalized coefficient second-moment bounds can be imposed jointly with this success event. On a polynomial-size realization, every active tensor degree, averaging count, and Poisson product length is polynomial. Active branches of inverse-superpolynomial probability may be excluded by an expected-call union bound; equivalently use the weighted product-coefficient estimate above. Thus, with positive probability, all normalized coefficients have magnitude at most $\exp(\mathrm{poly}(R))$. Initial normalized cube leaves are bounded signed coordinate/constant

formulas, and the final root RMS lies between the margin and a polynomial readout bound. No original control coefficient or internal RMS scale has to be encoded separately.

Fix this realization. A polynomial-size arithmetic tree with coefficient magnitudes $\exp(\mathrm{poly}(R))$ has values and coefficient derivatives bounded by the same form on Boolean inputs. Dyadic rounding to polynomially many bits therefore changes every group output by less than $1/4$, preserving the signs of the good three quarters at every input and hence the majority. Clear positive subtree denominators. Arithmetic-formula balancing yields polynomial-size logarithmic-depth integer formulas. If complex gates were retained, first balance using their real-affine one-hole contexts, then expand real and imaginary parts; this also gives only polynomial growth. Formula positivity is in PNC^1. The composition theorem for PNC^1 predicates absorbs the final majority [1, Theorem 4.2], and $\mathsf{PNC}^1 \subseteq \mathsf{L/poly}$ in the nonuniform setting follows from [1, Proposition 2.4].

6 Entropy budgets and the Gaussian scalar baseline

In the actual Gaussian path, take $A_j = \Sigma_j^{1/2}$ in the current fixed feature basis. The scalar inequality $(\sqrt{\lambda} - 1)^2 \le \lambda - 1 - \log \lambda$ gives

$$\kappa_j \le \sqrt{2D_j}, \qquad M_j \le \sqrt{2D_j}.$$

A one-dimensional entropy marginal gives

$$\log_+ \left\| A_j^{-1} \right\|_{\mathrm{op}} \le C(\sqrt{D_j} + D_j).$$

Consequently $S + L \le C \sum_j (\sqrt{D_j} + D_j)$. The kernel semantics are exact throughout: positive covariance square roots act on the existing feature space, and the noncentral Hermite expansion realizes (1) including all query-query inner products. The final physical norm is at most one, so dividing the final feature by its positive query norm preserves the output sign and cannot reduce a unit margin. This proves Theorem 1.1.

For $q > 0$, let $V_G(q)$ be the infimum over finite scalar paths, including length zero, of

$$\sum_{j=1}^{m} \mathrm{KL}(\eta_j \| N(0, q_{j-1})) + \frac{1}{8q_m}, \qquad q_0 = q, \quad q_j = \mathbb{E}_{\eta_j} \tanh^2 Z,$$

where every η_j is Gaussian, with arbitrary mean and positive variance. Write $V(q)$ for the unrestricted scalar value and $V_* := \lim_{q \downarrow 0} V(q)$.

Proposition 6.1 (Different scalar baselines)**.** *The limit $V_*^G := \lim_{q \downarrow 0} V_G(q)$ exists, and there are universal $C_0 < \infty$ and $\delta_0 > 0$ such that*

$$V_* + \delta_0 \le V_*^G < 2, \qquad 0 \le V_*^G - V_G(q) \le C_0 q \quad (0 < q \le 2).$$

Proof. Scaling all successive controls proves that V_G is nonincreasing; Gaussianity and relative entropy are preserved. Let $F(u) = \mathbb{E} \tanh^2(\sqrt{u} Z)$. The bound $F(u) \ge u/(1 + 3u)$ shows that Gaussian variance inflation from q to $(1 + 6q)q$, for $q < 1/10$, costs at most $9q^2$ and increases the feature variance by at least $q^2(3 - 18q)/(1 + 3q + 18q^2)$. Integrating the cost-to-increment ratio, and shortening the last step to hit $1/10$, gives total cost including readout at most

$$U = 199/200 + \log(5/2) < 2.$$

Thus the limit exists and is at most U. Likewise $F(u) \ge u(1 - 2u)$ shows that inflation to $(1 + 16q)q$ costs at most $64q^2$ and advances the state by at least $4q^2$ for $q \le 1/128$. Growing from an arbitrarily

small state to $p \le 1/128$ costs at most $17p$. Appending a Gaussian suffix proves $V_*^G - V_G(p) \le 17p$; boundedness extends this to $p \le 2$.

To prove strict separation, put $b = 1/16$, $a = b/10$, and

$$I_p(r) = \sup_{\theta \in \mathbb{R}} \{\theta r - \log \mathbb{E}_{N(0,p)} e^{\theta \tanh^2 Z}\}.$$

For $0 < r < 1$, the unique entropy-minimizing law with $\mathbb{E} \tanh^2 Z = r$ has density proportional to $e^{\theta \tanh^2 Z}$ relative to $N(0,p)$. Its entropy $I_p(r)$ is continuous in (p,r). Consider Gaussian laws $N(m,s)$ with $p \in [a,b]$, entropy at most two relative to $N(0,p)$, and output $r = \mathbb{E}_{N(m,s)} \tanh^2 Z \ge b$. Their parameter set is compact: the entropy bounds m^2/p and confines s/p to a compact subinterval of $(0,\infty)$. Their outputs are bounded away from one. On this set the continuous entropy saving

$$\mathrm{KL}(N(m,s) \| N(0,p)) - I_p(r)$$

has a positive minimum δ_0. Indeed equality would make the Gaussian log-density ratio a bounded function $\theta \tanh^2 z - \log \mathbb{E} e^{\theta \tanh^2 Z}$. A quadratic polynomial can be bounded on $\mathbb{R}$ only if it is constant, forcing the reference Gaussian and hence $r = F(p) < p \le b$, a contradiction.

Any Gaussian scalar path of cost below two from $q < a$ must cross b, since otherwise its readout costs at least two. At its first crossing, the preceding state p is at least a: a post/pre-state ratio of at least ten forces entropy at least $(10 - 1 - \log 10)/2 > 2$ by Gaussian second-moment entropy duality. Replacing that single Gaussian law by the optimizer defining $I_p(r)$ preserves the post-state and saves at least δ_0. Hence $V_G(q) \ge V(q) + \delta_0$. Paths of cost below two approximate $V_G(q)$ by the upper bound $U < 2$; taking $q \downarrow 0$ proves the claim. □

Lemma 6.2 (Occupation relative to the Gaussian baseline)**.** *For a globally* $1/2$*-margined Gaussian path with action* $A = \sum_j D_j + \|v\|^2/2 \le V_*^G + \epsilon$*,* $\epsilon \ge 0$*,*

$$\sum_j \sqrt{D_j} \le C[1 + \log(H+2) + \sqrt{H\epsilon}], \qquad \sum_j D_j \le U + \epsilon.$$

Moreover every such path, without the upper-action hypothesis, satisfies $A \ge V_*^G - C/H$*.*

Proof. Let $q_i = \max_x \|h_i(x)\|^2$. Projection of a Gaussian row law onto any input is Gaussian, and data processing bounds its entropy by D_j. The scalar terminal readout bound is $1/(8q_H(x))$. Therefore a scalar Gaussian suffix from an input attaining q_i gives

$$S_i := \sum_{j \le i} D_j \le A - V_G(q_i) \le \epsilon + C_0 q_i.$$

Strict scalar dissipation gives $D_j \ge q_j - q_{j-1} + c q_{j-1}^2$ for a universal $c > 0$. One direct proof uses $\log \mathbb{E} e^{\tanh^2(\sqrt{p}Z)} \le p - cp^2$ on $[0,2]$: the heat equation makes the inequality strict away from zero, and the expansion is $p - p^2 + O(p^3)$ at zero. For maximum variances, decrease c until $p - cp^2$ is increasing. These facts also handle $q_0 \le 2$.

On an interval with pre-states and endpoint at least u, entropy E and length ℓ, divide dissipation by the squared pre-state to get $\ell \le C(E/u^2 + 1/u)$. Use last exits from dyadic levels starting at $\max\{\epsilon, (H+1)^{-1}, \min_i q_i\}$. Each later band below a fixed positive level has entropy $O(u)$ and length $O(1/u)$, so Cauchy–Schwarz bounds its square-root entropy by a constant. There are $O(\log(H+2))$ bands. The initial prefix has either entropy $O(\epsilon + 1/H)$ and length at most H, or entropy $O(b)$ and length $O(1/b)$ with $b = \min_i q_i$; its contribution is $O(1 + \sqrt{H\epsilon})$. The remaining high-variance tail has bounded length. This proves the claim for $\epsilon < 1/4$; otherwise apply Cauchy–Schwarz directly to $\sum D_j \le U + \epsilon$.

For the lower-action assertion it suffices to consider $A \le V_*^G$. Then $S_i \le C_0 q_i$ and $\sum D_j < 2$. The same last-exit argument, starting at $b = \min_i q_i$, bounds the total length by C/b. Thus $b \le C/H$, and its Gaussian suffix costs at least $V_G(b) \ge V_*^G - C_0 b$. □

Corollary 6.3 (Near the Gaussian class's own baseline)**.** *Suppose a globally* $1/2$*-margined Gaussian-controlled path has action at most* $V_*^G + \epsilon$*. Its classifier has* PNC^1 *resources polynomial in*

$$(d + H + \epsilon + 2) \exp\Big[C\{1 + \log(H+2) + \sqrt{H\epsilon} + \epsilon\}\Big].$$

In particular, at $\epsilon = O(\log^2 R/H)$ *with* $d, H \le \mathrm{poly}(R)$ *and* $H \ge c \log R$*, these resources are polynomial in* R*.*

Proof. Scale the margin to one and apply Theorem 1.1 and Lemma 6.2. The action also bounds the readout norm by $O(\sqrt{1+\epsilon})$. □

Proposition 6.4 (Attainability from the prescribed input)**.** *Let* $J_H^G(f)$ *be the Gaussian-restricted action infimum for global margin* $1/2$*, and let* Q_1 *be the first free tanh kernel. Set*

$$a_f = (f^T Q_1^{-1} f)^{-1}, \qquad S_f = 1 + \log(1/a_f).$$

There is a universal C *such that*

$$V_*^G - C/H \le J_H^G(f) \le V_*^G + C\frac{S_f^2 + \log^2(2H)}{H} \qquad (H \ge C S_f).$$

One has $S_f \le 15(d+1)$*, while constants and dictators satisfy* $S_f = O(\log(d+1))$*. In particular every fixed target has deep Gaussian rate limit* V_*^G*, and the own-baseline corollary is nonempty in its polynomial-resource range.*

Proof. The lower bound is Lemma 6.2. For the upper bound use the target-component construction of [2]. We check specifically that its laws remain Gaussian. The first free kernel is positive definite and dominates $a_f f f^T$. Free cooling, holding, prescribed growth, and purification use only Gaussian covariance inflations in this target direction. A smaller tracked component merely chooses a new decomposition $Q = v f f^T + R$, $R \succeq 0$; it changes no law. Representing the reference as $\sqrt{v} f Z + G_R$, with independent Gaussian terms, shows by data processing that each scalar Gaussian modification induces a Gaussian vector control with no greater entropy, even for singular R.

The increasing-path reduction and small-state bridge in Proposition 6.1 preserve Gaussianity. The usual dyadic entropy bound therefore gives a Gaussian scalar suffix from p with cost at most $V_G(p) + p$ and length $O(1/p)$. Its stable lift rescales this Gaussian scalar law and adds an independent residual Gaussian, so it too is admissible. Choosing cooling scale $\eta \asymp S_f/H$ and target scale $p \asymp [S_f + \log(2H)]/H$, the construction has excess action at most

$$C[H\eta^2 + S_f\eta + p + p\log(2H) + H^{-4}] \le C[S_f^2 + \log^2(2H)]/H.$$

Exact Gaussian holding fills the unused layers before growth. The first-layer spectral bound gives $a_f \ge e^{-14(d+1)}$; its first Hermite term gives $a_f \ge c/(d+1)$ for constants and dictators. These are the asserted bounds on S_f. □

Proposition 6.1 and Lemma 6.2 show why the choice of baseline matters: the Gaussian subclass with action at most $V_* + \epsilon$ is empty whenever $\epsilon < \delta_0 - C/H$. A near-$V_*$ Gaussian consequence would therefore be asymptotically vacuous. The unrestricted posterior admits all finite-entropy row tilts. Neither a Gaussian replacement theorem nor a formula compiler for those tilts is proved here, so the original learning question remains open.

References

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[2] *Critical depth and learning*. Research manuscript, sections "Scalar control and target-component constructions", "Short near-optimal scalar paths and residual spectral control", and "The initial target component and the sufficient depth". https://github.com/kaarelh/AI-works/tree/main/neural-network-generalization/works-in-progress/critical-depth.